PT VECTOR MANAGEMENT CONSULTING

PROPOSAL FOR BUILDING

Dynamic Operational Synchronization & Scheduling Platform

FOR

Berau Coal Transshipment Operations

Prepared for	ABL Group
Prepared by	PT Vector Management Consulting
Operating context	ABL - Berau Coal transshipment, port, barging, CTS and OGV loading operations
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1. Executive Summary

ABL and Berau Coal operate in a highly dynamic and deeply interconnected coal transshipment environment involving cargo planning, coal quality and blending requirements, jetty loading operations, tug-and-barge movement, river navigation constraints, tide windows, bridge crossing windows, CTS availability, OGV loading commitments and field-level operational execution.

Managing transshipments planning and manual execution leads to issues like missing out on OGV scheduled dates while having underutilized assets.

ABL is considering an Auto Scheduling Software which can help them to plan their assets for maximum utilization and achieving an very high On Time and In full (OTIF) for the customers of Berau Coal.

ABL has requested Vector to submit a proposal for the same.

2. Current Operating Realities

- Planning is highly dynamic and frequently revised monthly, weekly, daily and sometimes intra-day.
- Demand and voyage planning are often created independently from real-time logistics capacity realities.
- Operational decisions depend heavily on manual coordination, spreadsheets, WhatsApp and planner experience.
- Route cycle times vary significantly across jetties, creating uneven effective fleet capacity.
- Tide windows, bridge schedules and navigation constraints continuously impact movement feasibility.
- Asset positions may be visible through AIS/GPS systems, but operational business-state visibility remains limited.
- CTS allocation, tug-barge reassignment and OGV changes create continuous replanning requirements.
- Field teams often make critical decisions without structured simulation-backed support.
- Current KPIs may drive local optimization instead of synchronized end-to-end flow performance.
- Operational disruptions propagate rapidly across the entire chain due to lack of synchronized visibility.

3. Root Cause Perspective

The current challenge is not just a scheduling software problem. Bigger Issue is, Berau Coal generates demand commitments and OGV plan without continuous synchronization with

- Actual WIP,
- Tug-barge loading,
- Route cycle realities,
- Tide windows,
- Bridge constraints,
- CTS capacities,
- Jetty congestion,
- Real operational availability,

The above resulting in operational mismatch and instability. Operation therefore suffers from

- Reactive planning,
- Excessive waiting,
- Underutilized assets,
- Repeated reassignment

The fundamental business need is therefore:

"Synchronizing demand commitments with executable operational flow capacity."

4. Vector Recommendations

The proposed product must be **Dynamic Operational Synchronization & Scheduling Platform**. It should convert Berau shipment demand into an executable ABL logistics plan by coordinating cargo readiness, jetty loading, tug-barge movement, tide and bridge windows, CTS availability, OGV loading sequence, exception recovery, approval and publication.

The platform should not only be a fleet-tracking screen, map view or digital Excel replacement. It should be a governed scheduling and simulation application designed to place each shipment movement through eligible route steps, available operating windows and approved decision workflows.

Vector will use its proprietary scheduling, scenario and dynamic-constraint scaffolding as the base for the application build. The ABL-specific implementation will configure and extend this base around Berau demand, ABL assets, tide and bridge logic, port/CTS operations, recovery workflows, approval governance and required integrations.

5. Business Use Case

Operating principle: Synchronize Berau Coal's shipment commitments with ABL's real operational flow capacity so that cargo readiness, asset availability, tide/bridge windows, jetty/CTS capacity, OGV loading sequence and recovery decisions are planned and governed from one shared operating platform.

5.1 Business problems addressed

Business problem	Application response
Static Excel planning cannot absorb daily and intra-day changes	Structured demand intake, versioned plan generation, conflict checks, approval and governed replanning
Shipment demand is not continuously synchronized with logistics capacity	OGV demand, cargo grade, route, tug-barge, jetty, CTS and operating-window capacity are held in one planning model
Tide, bridge and berth windows keep changing the feasible plan	Movement windows are treated as schedulable capacity
Coal grade and loading sequence affect execution	Cargo requirement and layer sequence are first-class planning inputs
Existing work in progress affects new planning	Active plans, sailing barges, loaded barges, queues and confirmed actuals are plotted before new planning decisions
Operating exceptions are currently handled through experience and manual coordination	Exception families, reason codes, manual override and recovery workflows are captured in the governed process
Asset position is visible but operational status may require confirmation	GPS/AIS evidence is separated from confirmed operational events and schedule authority
Approvals and operating decisions require traceability	Dual-party approvals, immutable published snapshots, audit events and governed exports

6. Core Scheduling Philosophy

Operating principle: Each shipment movement is treated as a schedulable entity that must pass through a defined route of operational steps. Each step needs the right resource, enough available capacity and a valid operating window.

For ABL, the route is operational as well as physical:

```
Cargo readiness
-> Jetty / BLC loading
-> Tug-barge departure
-> Tide / river movement
-> Bridge crossing
-> CTS / anchorage queue
-> CTS discharge
-> OGV hatch / layer completion
-> Survey / handover closure
```

The controlling constraint shifts during the day between tide, bridge, jetty, CTS, berth, tug-barge availability, cargo readiness and OGV laycan pressure. The application will therefore recalculate feasibility around moving operating windows and current execution state, not only around static planned dates.

7. Vector Proprietary Scheduling And Scenario Scaffolding

Vector owns a proprietary base scaffolding for constraint-aware scheduling, scenario simulation and dynamic-capacity planning. This base will be used as the starting point for the ABL application build.

7.1 Capabilities provided by the base scaffolding

Capability	Role in ABL build
Scheduling entity model	Represents shipment movements, route steps, dependencies, priorities and target dates
Route and resource eligibility	Matches each movement with allowed jetties, routes, tug-barge pairs, CTS assets and operating steps
Capacity-window placement	Places movements into free operating windows and prevents invalid overlap
Current-state loading	Considers active work, occupied resources, queues and confirmed actuals before placing new work
Dynamic constraint evaluation	Rechecks feasibility when tide, bridge, berth, asset or cargo readiness changes
Scenario engine	Simulates delay, outage, window-change, reassignment and recovery assumptions
Candidate comparison	Evaluates alternate paths and recovery options against feasibility and impact

Capability	Role in ABL build
Exception and blocker output	Identifies the first operating blocker and the affected downstream commitments

7.2 ABL-specific configuration and extension

The ABL build will configure this base around:

- Berau OGV demand, cargo quantity, coal grade and layer sequence
- ABL jetties, BLC loading capability, tug-barge fleet and CTS assets
- tide windows, bridge windows, river movement and route cycle times
- cargo readiness, stockpile/source eligibility and quality-release status
- active trip state, loaded barges, queues and confirmed operational events
- exception categories, recovery rules, approvals and publishability gates
- reporting, dashboard, audit and export requirements.

The proprietary base accelerates implementation while the blueprinting phase ensures that ABL-specific business rules, data interfaces and governance logic are validated before full deployment.

8. Product Development Layers

8.1 Layer 1 - Governed scheduling model

This layer creates the structured planning spine:

- OGV voyage and laycan intake
- cargo quantity, coal grade and loading sequence
- source, stockpile, jetty and route master data
- tug, barge and CTS availability
- tide and bridge calendars
- deterministic schedule generation
- conflict detection
- manual override with reason capture
- plan versioning
- approval, publish and audit.

8.2 Layer 2 - Operating readiness and exception governance

This layer organizes how the scheduling model is adopted into day-to-day planning:

- planner workflow validation
- demand completeness and master-data readiness
- conflict reason review
- override pattern review
- exception family definition
- approval latency review
- readiness review before activating later operating layers.

8.3 Layer 3 - Simulation and operational evidence

This layer adds scenario and live-state interpretation:

- delay, outage, window-change and reassignment assumptions
- projected trip, event, OGV completion and utilization impact
- GPS/AIS-style position evidence
- geofence, ETA variance and stale-signal alerts
- event candidates from device, operator or feed input
- trusted confirmation before schedule actualization.

8.4 Layer 4 - Recovery and guided execution

This layer adds guided decision support:

- recovery input snapshots from active plan, alerts, confirmed events and conflicts
- ranked recovery options
- root-cause validation
- scenario creation from selected recovery option
- publishability assessment
- guided operator workflow
- projection-only commercial and operating-impact review surfaces.

9. Phase Of Development in Sprint Level

Sprint level	Build focus	Key outcome
Sprint 0	Application foundation: web application, API service, database, background processing, file storage and routing layer	Runnable product spine
Sprint 1	Master data and governance: organizations, roles, audit, locations, cargo, jetties, routes, assets and compatibility	Safe multi-party operating model
Sprint 2	Demand and capacity intake: OGV demand, cargo layer, asset windows, jetty windows, tide windows and bridge windows	Structured inputs for planning
Sprint 3	Scheduling entity and route model: trips, assignments, schedule events and route-step dependencies	Movement chain represented as schedulable work
Sprint 4	Capacity placement and feasibility: candidate paths, occupied/free windows, conflict checks and movement candidates	Feasible plan with explicit blockers
Sprint 5	Plan governance: versions, overrides, approval requests, approval decisions, published snapshots and exports	Approved execution contract
Sprint 6	Scenario simulation and exception review: assumptions, projected events, projected trips, constraint evaluations, exception families and utilization outputs	What-if planning and exception impact visibility
Sprint 7	Live evidence: telemetry sources, asset identities, position pings, latest asset state, geofences and ETA variance	Planned-vs-observed movement visibility
Sprint 8	Confirmed operations: integration feeds, devices, event candidates, confirmed events, actualization and device health	Trusted execution status layer
Sprint 9	Recovery recommendations: input snapshots, ranked options, recovery actions and proof pack	Decision support for disrupted flow

Sprint level	Build focus	Key outcome
Sprint 10	Guided recovery and hardening: workflow guidance, root-cause validation, publishability, review surfaces and tests	Pilot-ready governed flow-management platform

The first release will prove the governed scheduling spine and operating-window logic. Later releases will progressively add simulation, live evidence, trusted field events, recovery guidance and management review surfaces.

9.1 Client delivery blocks

Delivery block	Sprint coverage	System screens delivered	Capability delivered	Product owned by ABL at this stage
Release 1 - Governed scheduling spine	Sprint 0 to Sprint 5	Login and role-based navigation, master data, OGV demand, cargo layer sequence, tide/bridge windows, tug-barge assignment, jetty loading, CTS operations, conflict review, override reason capture, approval, published plan, audit and export screens	Structured demand intake, master-data setup, operating-window entry, route/resource eligibility, candidate movement generation, feasibility checks, conflict output, plan versioning, dual approval, manual publish and governed export	A working governed scheduling application that replaces spreadsheet-only planning for the core ABL-Berau flow and creates an approved execution contract
Release 2 - Exception and scenario support	Sprint 6	Exception review, scenario assumption entry, baseline-versus-scenario comparison, projected trip/event impact, constraint impact and utilization review screens	Exception family handling, what-if planning for delay/outage/window change/reassignment, projected downstream impact and scenario promotion into approval path	A planning simulation and exception review product that allows ABL to test disruption impact before changing the published plan
Release 3 - Live evidence and confirmed operations	Sprint 7 to Sprint 8	Live resource map, asset signal detail, tracking alert list, operations event console, device/feed health, jetty/CTS actualization and exception linkage screens	GPS/AIS-style evidence ingestion, latest asset state, geofence/ETA variance, stale-signal alerts, event candidate capture, event confirmation, actual execution update and trusted exception creation	An execution-monitoring layer that connects the approved schedule with observed and confirmed field reality
Release 4 - Recovery guidance and flow-management hardening	Sprint 9 to Sprint 10	Recovery recommendation console, ranked option detail, root-cause validation, publishability review, guided workflow	Ranked recovery options, recovery proof pack, guided next action, publishability validation, telemetry-trust review and projection-only	A governed flow-management platform that supports disruption recovery from exception identification to

Delivery block	Sprint coverage	System screens delivered	Capability delivered	Product owned by ABL at this stage
		panel, management review surfaces and hardened audit evidence screens	operating/commercial visibility	approved recovery publication

10. Release 1 - Governed Scheduling Spine

Release 1 establishes the governed scheduling spine and operating-window logic across Sprint 0 to Sprint 5. It creates the first complete product layer that ABL can use for structured planning, feasibility review, approval and publication.

10.1 Release 1 system screens

Screen group	Release 1 coverage
Access and governance	Login, role-based navigation, organization, user, role and approval authority screens
Master data	Locations, cargo grades, sources, stockpiles, jetties, routes, tug/barge/CTS assets and compatibility
Demand and cargo	OGV demand intake, laycan, cargo quantity, coal grade and cargo layer sequence
Operating windows	Asset availability, jetty/BLC windows, tide windows, bridge windows and CTS capacity windows
Scheduling	Candidate movement generation, tug-barge assignment, schedule events, conflict output and draft plan review
Approval and publish	Override capture, approval request, approval decision, publishability check, published snapshot and export
Audit and evidence	Plan history, approvals, manual overrides, exports and audit events

10.2 Release 1 capability delivered

Release 1 delivers:

- structured demand intake
- master-data and resource setup
- cargo layer and loading sequence representation
- route and resource eligibility checks
- operating-window entry and validation
- candidate movement generation
- explicit conflict and blocker output
- reason-coded manual override
- plan versioning
- dual approval
- published plan snapshot
- governed export and audit trail.

10.3 Release 1 operating success measures

Success measure	Validation direction
Publishable-plan cycle time	Time from demand intake to publishable plan becomes visible and measurable
Schedule rework	Repeated spreadsheet revisions and manual reconciliation are reduced
Conflict clarity	Hard blockers carry a reason code, affected movement and review action
Approval traceability	Published plans carry approval, version and audit lineage
Planner adoption	Named users can complete representative planning workflows with controlled override usage
Data readiness	Demand, master data and operating-window inputs meet agreed minimum completeness

Specific numeric targets will be baselined with ABL and Berau during pilot use.

11. Operational Readiness And Adoption Review

The operational readiness review confirms that the governed scheduling spine is aligned with real planning behavior before subsequent product layers are activated.

The review will cover:

- planner usage across representative scheduling workflows
- demand completeness and master-data stability
- conflict accuracy and false-positive blockers
- manual override frequency and reason-code quality
- approval latency and escalation behavior
- readiness of operating-window inputs
- exception families that need formal workflow treatment
- data and integration gaps to be resolved before live evidence or recovery automation.

The output of this review becomes the operating baseline for scenario simulation, exception support, live evidence, confirmed operations and recovery guidance.

12. Exception Taxonomy And Operating Governance

ABL transshipment execution requires a practical exception structure because disruptions can occur at multiple points in the flow. The application will treat exceptions as governed operating events, not only as informal planner notes.

12.1 Initial exception families

Exception family	Initial examples	Application treatment
Cargo/source readiness	Cargo not ready, quality hold, source switch, cargo layer mismatch	Readiness conflict, planning note or manual resolution
Jetty/BLC operations	Loading delay, queue, reduced loading rate, equipment stoppage	Capacity-window conflict or loading exception

Exception family	Initial examples	Application treatment
Navigation constraint	Tide missed, bridge window missed, draft restriction change, crossing risk	Operating-window blocker
Fleet/asset availability	Tug unavailable, barge unavailable, incompatibility, breakdown, charter lead-time pressure	Resource availability or eligibility blocker
CTS/transshipment	CTS queue, floating crane downtime, discharge rate reduction	Downstream capacity blocker
OGV/laycan	OGV ETA change, hatch sequence change, laycan pressure, loading completion risk	Demand priority and completion-risk note
Weather/marine	Rain, wind, visibility, wave restriction, river condition	Exception note or window blocker
Survey/clearance	Draft survey delay, documentation hold, port clearance interruption	Closure, readiness or handover blocker
Human/governance	Approval delay, manual override, escalation, disputed decision	Workflow and audit event
Data/integration	Missing feed, stale signal, duplicate event, untrusted telemetry	Data quality warning

12.2 Manual override and planner discretion

The application will preserve planner discretion where practical operating judgment is required. Manual overrides, negotiated exceptions and escalation decisions will remain possible, but they will be captured with reason code, user, timestamp, affected movement and approval lineage.

Repeated manual decisions can then be reviewed with ABL and Berau to decide whether they should become formal scheduling rules, exception categories or operator guidance.

13. Working Flow Details

13.1 Clean planning flow

```
Import OGV demand
-> Review cargo grade and layer sequence
-> Enter operating windows
-> Generate movement candidates
-> Generate draft plan
-> Review conflicts
-> Apply reason-coded override if required
-> Submit approval
-> Complete Berau / ABL approvals
-> Run publishability check
-> Manually publish
-> Generate governed export
```

13.2 Disruption and recovery flow

```
Open active exception
-> Review operating cause
-> Generate recovery options
-> Review ranked candidates
-> Validate whether the option resolves the operating cause
-> Create scenario from selected option
-> Run simulation
-> Promote feasible scenario
-> Repair remaining blockers if required
-> Submit approval
-> Complete approvals
-> Run publishability check
-> Manually publish recovery plan
```

13.3 Capacity placement logic

At planning time, the platform will:

1. load current demand and master data
2. plot current work and already occupied capacity
3. identify eligible resources for each movement
4. generate candidate paths
5. search for continuous feasible windows
6. mark selected windows as occupied
7. record conflicts where no feasible placement exists
8. preserve reason codes and audit lineage.

13.4 Live evidence and actualization

```
GPS/AIS or synthetic position evidence
-> Normalize asset identity
-> Store raw evidence
-> Derive latest state, geofence and ETA variance
-> Raise alert where required
-> Use alert as scenario input when replanning is needed
```

```
Device/operator/event candidate
-> Match to planned trip or schedule event
-> Apply confidence, trust and dedupe rules
-> Confirm or reject
-> Update actual event only after confirmation
-> Create exception if variance requires action
-> Record audit trail
```

11.5 Operating state separation

State	Meaning
Planned	Approved schedule expectation
Observed	Field signal or external evidence that may indicate status
Confirmed	Operationally trusted event that updates actual execution status
Projected	Future impact estimated by the scheduling or scenario engine
Recommended	Candidate action for review, not an automatic publication

12. Application Modules

Module	Working responsibility
Organizations and access control	Multi-party users, roles, permissions, data scope and approval authority
Master data	Locations, coal grades, mines, stockpiles, jetties, tug/barge/CTS assets, routes, loading rates and compatibility
Planning	OGV voyages, cargo requirements, cargo layer sequence, availability windows, tide windows, bridge windows and import jobs
Scheduling	Plan, plan version, trip, assignment, schedule event, conflict, override, approval, publish, export and simulation
Exception management	Exception family, reason code, operating note, escalation path and recovery linkage
Telemetry	Source registry, asset mapping, position evidence, geofence zones, latest state, ETA projection and tracking alerts
Operations	Integration feeds, device endpoints, health snapshots, event candidates, confirmed events and actualization
Guided workflow	Operator guidance, action enablement and flow evidence
Audit	Request and domain audit trail for governed actions
Operator cockpit	Dense operational cockpit covering demand, cargo sequence, assignments, jetty, CTS, tide/bridge, exceptions, simulation, approvals, published plan, map, recovery, audit and admin

13. Assumptions Made To Determine The Flow

1. Berau provides OGV demand, laycan, cargo quantity, coal grade and layer sequence in a consistent file or integration-ready format.
2. ABL and Berau will agree master-data naming for jetties, routes, assets, grades, locations and operating zones.
3. Tide and bridge windows are initially entered manually or uploaded before live sensor integration is available.
4. Resource capacity includes both time availability and operational eligibility.

5. Active work in progress is plotted before new schedules or recovery options are generated.
6. GPS/AIS or Spinergie-style fleet data is evidence and does not directly change schedule authority.
7. Field events such as loading start, loading completion, bridge crossing, tide gate pass and CTS discharge require confirmation before actualization.
8. Published plans are immutable; replanning creates successor versions.
9. Approval requires Berau and ABL authority before publication.
10. Manual override requires reason capture and audit.
11. Recovery recommendations do not bypass simulation, approval or manual publish.
12. External chartered assets require readiness and lead-time assumptions.
13. Commercial exposure is limited to planning projection, while final demurrage, despatch, invoicing and settlement remain separately scoped.
14. Synthetic seed and replay data are valid for blueprint, prototype, UAT and integration-contract proof before production feeds are connected.

14. Proposed Implementation Architecture

The implementation architecture provides the operating foundation behind the planner and approval workflows.

Layer	Technology	Role in the application
Operator web application	React, Vite, TypeScript	Browser-based screens for planners, approvers and operations users
Web application testing	Vitest, Testing Library, jsdom	UI behavior, screen logic and component interaction testing
API and business services	Django, Django REST Framework	Business APIs, workflow rules, access control and application services
Background jobs	Celery, Celery Beat	Imports, exports, scheduled checks and periodic processing
Application server	Gunicorn	Production application serving pattern
Service testing	pytest, pytest-django	Business services, data rules, API behavior and scheduling logic testing
Database	PostgreSQL with PostGIS	Operational data store with geospatial capability where required
Cache / broker	Redis	Fast shared state and message brokering for background jobs
Object storage	MinIO / S3-compatible storage	Imported files, exports, generated documents and object artifacts
Edge routing	Nginx	Web/API routing, reverse proxy and deployment hardening
Runtime approach	Containerized deployment model	Repeatable deployment units for local, UAT and production environments

Deferred integration technology includes, where required by confirmed integration scope, MQTT, HTTP callbacks, AIS/NMEA feeds, Kafka/Redpanda, TimescaleDB or ClickHouse, OPC-UA or Modbus, and live UI push updates.

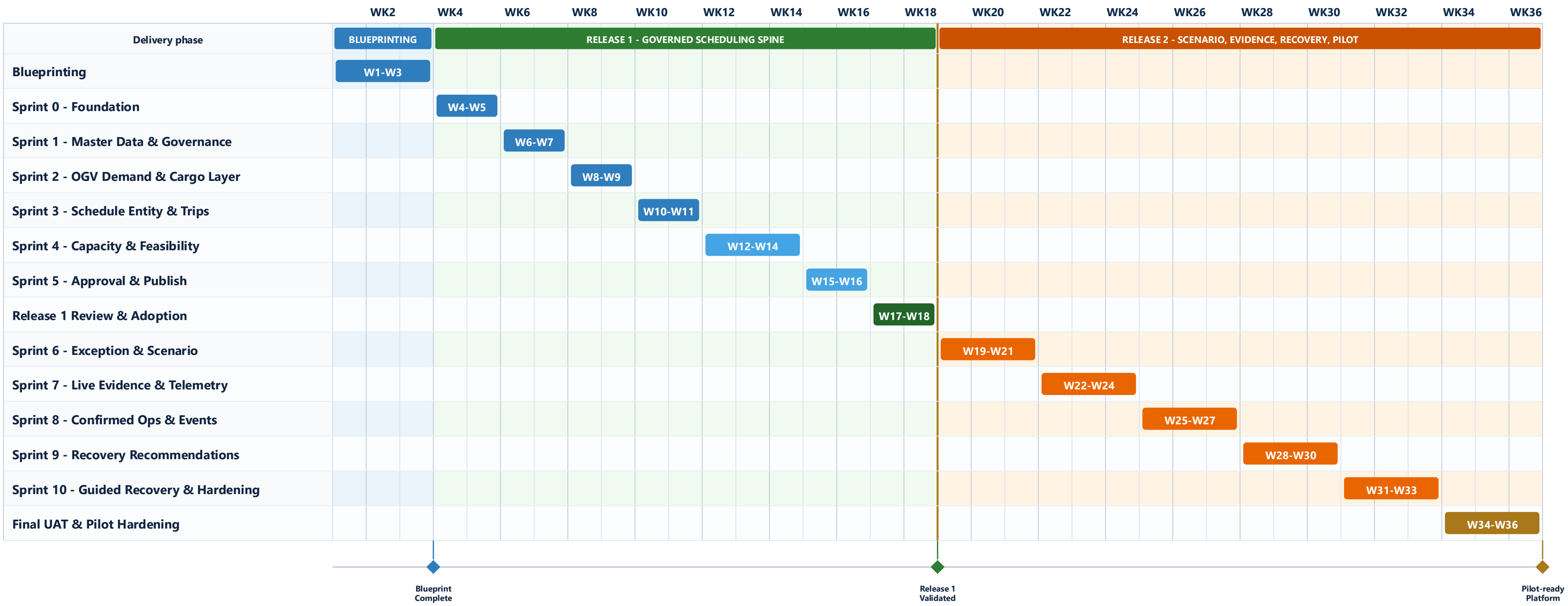
15. External Data Sources Assumed To Be Available

External data source	Assumed content	Initial handling	Later integration direction
Existing Excel schedule	Current OGV plan and operating schedule	Import or manual upload	Governed file ingestion or direct integration
OGV schedule	ETA, ETB, laycan, quantity, priority and vessel status	Manual/imported demand	API feed from Berau or shipping system
Cargo/source data	Mine, CPP, stockpile, grade, product, available quantity, quality release and jetty eligibility	Master data and demand setup	Cargo/quality or ERP integration
Cargo layer sequence	Hatch/layer order, grade requirement and substitution rule	Cargo layer model	Structured demand feed
Jetty/BLC data	Loading rate, queue, working hours, equipment status and readiness	Manual windows/status	Terminal, PLC or operator console
ABL fleet data	Tug, barge, CTS capacity, compatibility, status, location and availability	Master data and availability windows	Spinergie, GPS/AIS or fleet API
Tide data	Tide tables, height, safe movement windows and water-level observations	Manual tide windows	Sensor/API feed
Bridge data	Opening schedule, crossing rule and queue	Manual bridge windows	Bridge operator console or device feed
Weather/marine data	Rain, wind, wave, visibility and closure alerts	Exception input or planning note	Weather API or local station
GPS/AIS data	Position, speed, heading, timestamp, signal quality and external asset identity	Synthetic replay or vendor-shaped payload	Spinergie, AIS receiver, tracker or vendor API
Operational events	Loading start/end, discharge start/end, stoppage, breakdown and crossing confirmation	Synthetic feed and operator confirmation	MQTT, HTTP callback, PLC, weighbridge, tablet or edge gateway
Device health	Last seen, power, battery, network, firmware and signal gap	Device health snapshot	IoT/device-management integration
Survey and documents	Draft survey, sampling, certificates, SOF and final quantity/quality documents	Deferred milestone visibility	Document management or survey integration
Commercial parameters	Laycan risk, waiting-time proxy, queue impact, utilization and exposure proxy	Projection-only review	Separate commercial sign-off

Program Delivery Timeline

ABL | Dynamic Operational Synchronization Platform

Estimated delivery span: 34-36 weeks from kickoff | Blueprinting: W1-W3 | Release 1 validation: W18 | Pilot-ready platform: W36



BLUEPRINTING
W1-W3
Operating rules, data assumptions, flow logic, and sprint backlog.

RELEASE 1
W4-W18
Governed scheduling spine, operating window logic, and pilot-user validation.

RELEASE 2
W19-W33
Scenario handling, live evidence, confirmed operations, and recovery guidance.

PILOT READY
W34-W36
Full UAT, pilot hardening, platform sign-off. External risk: W40-W44.

16. Build Deliverables

16.1 Release 1 deliverables

Deliverable	Description
Vector scheduling base configuration	ABL-specific setup of proprietary scheduling and dynamic-constraint scaffolding
Application foundation	Containerized web application, API service, database, Redis, object storage and routing layer
Master and capacity model	Assets, routes, eligibility, calendars, operating windows and compatibility
Demand and cargo intake	OGV demand, laycan, cargo quantity, coal grade and cargo sequence
Scheduling configuration	Shipment movement entity, candidate paths, capacity placement, moving-window checks and conflict output
Operator screens	Role-aware screens for demand, master data, windows, assignments, conflicts, overrides, approvals, publish and export
Governance layer	Access control, dual approval, publishability, immutable snapshots and governed exports
Evidence and test pack	Service tests, web application tests, proof commands, UAT scripts and runbook

16.2 Later release deliverables

Deliverable	Description
Exception and scenario configuration	Exception families, what-if assumptions, projected impacts, utilization views and scenario comparison
Recovery support	Recovery input snapshot, ranked options, impact review and proof pack
Telemetry-assisted execution	GPS/AIS-shaped evidence, latest state, geofence/ETA variance and data trust review
Operations event layer	Event candidates, confirmations, rejection, actual event updates and device/feed health
Management review surfaces	Exception patterns, operating impact, projection-only indicators and maturity reporting

16.3 Operating impact indicators

Where the required data is available, later release layers may expose projection-only operating indicators:

- laycan risk
- waiting time
- fleet utilization
- queue impact
- charter lead-time pressure
- demurrage exposure proxy.

These indicators support planning and review. Final demurrage, despatch, invoicing and settlement remain separately scoped.

17. Implementation Clarification

This extension defines the application build direction that follows the blueprinting and prototype scope.

Final implementation timeline, team size and commercials will be confirmed after:

- ABL and Berau validate the operational rules
- required external data sources are confirmed
- integration owners and access methods are known
- master-data ownership is agreed
- pilot users and approval authorities are named
- UAT scope and deployment environment are confirmed
- licensing and usage terms for Vector proprietary scheduling and scenario scaffolding are finalized.

18. Important Build Boundaries

1. GPS/AIS is evidence, not automatic truth.
2. Raw pings do not update the execution plan directly.
3. Confirmed field events update actual execution state.
4. Recovery options are advisory until simulated, approved and checked for publish readiness.
5. Published plans remain immutable.
6. External visibility is governed through approved publication rules.
7. Commercial projection is not final settlement logic.
8. Synthetic proof data is acceptable for prototype and UAT, but production claims require real integration testing.
9. Planned, observed, confirmed, projected and recommended states remain separate.

19. Glossary

Term	Definition
ABL	The operating organization responsible for transshipment logistics coordination in this proposal context.
AIS	Automatic Identification System, a marine tracking signal used to identify vessel location, heading, speed and related navigation information.
BLC	Barge Loading Conveyor or barge loading capability used at jetty/loading points for coal movement into barges.
Berau	Berau Coal, the shipment-demand context for the proposed ABL operating blueprint.
Bridge window	A permitted time range for a tug-barge movement to pass a bridge based on opening rules, navigation safety or operating coordination.
Candidate movement	A possible scheduled movement generated by the planning logic before it is accepted into a draft or published plan.
Cargo layer sequence	The required sequence in which coal grades or cargo layers are loaded into an OGV hold or hatch.
Constraint	A limiting factor that affects whether a movement can be planned, such as asset availability, tide timing, bridge opening, jetty capacity or cargo readiness.

CTS	Coal Transshipment Station or transshipment asset used for floating transfer/discharge operations between barges and ocean-going vessels.
Demurrage exposure proxy	A planning indicator that estimates potential commercial exposure due to waiting or delay. It is not final settlement logic.
Dynamic constraint	A constraint whose value or effect changes during operations, such as tide level, bridge availability, equipment status or asset position.
Feasibility check	A schedule validation step that confirms whether a movement can be placed within eligible resources and valid operating windows.
Geofence	A virtual geographic boundary used to detect whether an asset has entered, exited or remained within an operating zone.
Governed scheduling spine	The controlled planning workflow from demand intake through schedule generation, conflict review, approval, publish and export.
Jetty	The loading point where cargo is transferred to barges before movement toward CTS or OGV operations.
Laycan	The contractual or operating window within which an OGV is expected to be ready for loading.
Manual override	A planner action that allows a schedule decision to proceed despite a blocker or rule exception, with reason and audit capture.
MQTT	Message Queuing Telemetry Transport, a lightweight publish/subscribe protocol commonly used for IoT, device and event-message integration.
NMEA	National Marine Electronics Association data format family used by marine navigation equipment, often carrying GPS, AIS and vessel instrument data.
OGV	Ocean Going Vessel, the receiving vessel for the final coal-loading operation.
OPC-UA	Open Platform Communications Unified Architecture, an industrial communication standard used to exchange structured machine, equipment and process data.
Operating window	A time range within which an activity is permitted or feasible, such as tide movement, bridge crossing, jetty loading or CTS discharge.
Operator console	A local system, screen or control interface used by operational staff to enter, confirm or view equipment and process status.
PLC	Programmable Logic Controller, an industrial controller used to monitor and control equipment such as conveyors, loading systems and related plant operations.
Planned state	The approved schedule expectation before field execution changes are confirmed.
Publishability check	A final validation that a plan is ready to be released as an approved operating schedule.
Published plan	An approved, immutable schedule snapshot that becomes the operating reference until a successor version is approved.
Reason code	A standardized explanation attached to a conflict, exception or manual override.
Recovery option	A candidate corrective action proposed for review after an operational disruption.
Scenario	A what-if schedule view that tests assumptions before changing the published plan.
Tide window	A time range during which water level and navigation conditions allow a safe or permitted movement.

Tug-barge pair	A tug and barge combination considered together for movement eligibility, availability and compatibility.
UAT	User Acceptance Testing, the client-side validation stage for workflows, screens and outputs before production use.

20. Reference Inputs

This extension is based on:

- the Operational Blueprinting & Simulation Prototype Proposal
- the Scheduling Simulation BRD
- ABL and Berau Coal operating context
- observed transshipment planning and fleet-coordination challenges
- cargo, route, asset, tide, bridge, CTS and OGV scheduling logic
- the shared scheduling principle that work must be placed through eligible route steps, available resources and continuous operating windows
- the product-development direction for the ABL flow-management platform.

End of application build proposal

PT Vector Management Consulting